

Question 1:

Object:

Write a step-by-step algorithm to determine the year when a person will turn 100 years old.

Your algorithm must clearly list the steps for:

- Obtaining the three input values: name, current age, and current year.
- Calculating the year when the person will turn 100.
- Displaying a message addressed to the person that tells them the year they will
- turn 100.

ALGORITHM

Step 01: Start the Program.

Step 02: Declare Variables, Name, Age, current_year, Year_100,

Step 03: Read Name, Age, current_year.

Step 04: Calculate $\text{year_100} = \text{current_year} + (100 - \text{age})$.

Step 05: If $\text{age} < 100$ then write: "Hi {Name}, you will turn 100 in the year {year_100}."

Else if $\text{age} = 100$ then write: "Hi {Name}, you are 100 this year {current_year}."

Else write: "Hi {Name}, you turned 100 in the year {year_100}."

Step 06: Stop the program.